TIC 178434324

Eclipsing binary candidate dossier - Star Vetter

CLEAN

```
BLS EPHEMERIS
  Period        2.384441 d
  Epoch (t0)    1492.6977 BTJD
  Duration      0.1200 d
  Depth         26.266 %  (262,655 ppm)
  SNR           244.0

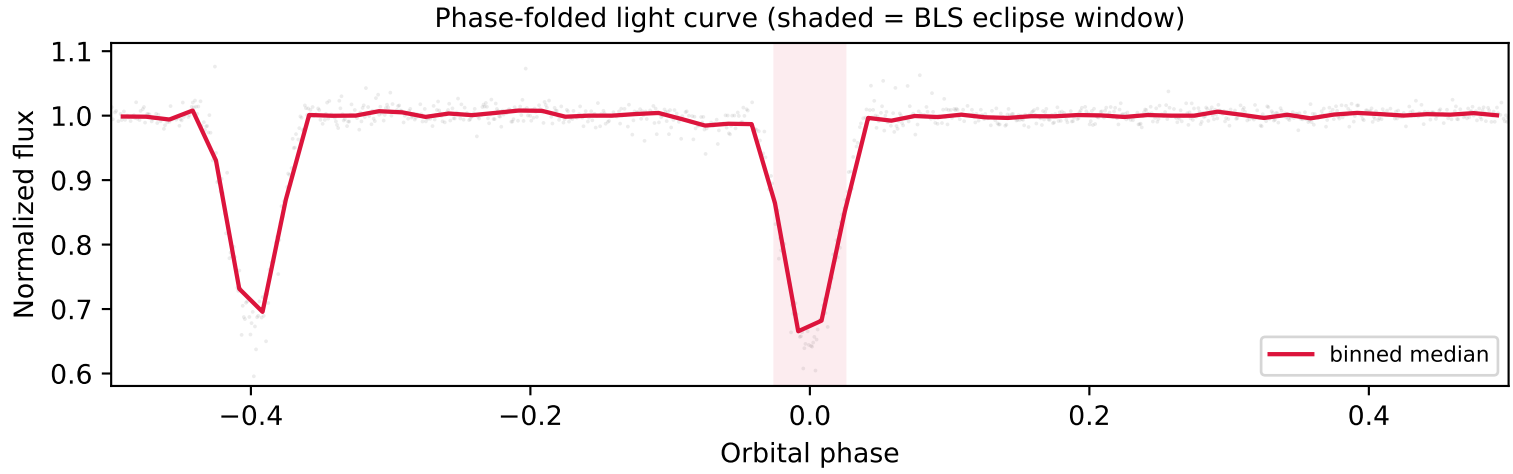
INDEPENDENT CHECKS (f3i gates)
  Eclipses observed      9  (gate >= 3)
  Robust depth ratio     1.11  (gate >= 0.50)
  Robust depth sigma     149.6  (gate >= 7.0)
  Odd/even z-score       2.12  (consistent)
  Secondary eclipse      detected, phase -0.40, 5.5 sigma
```

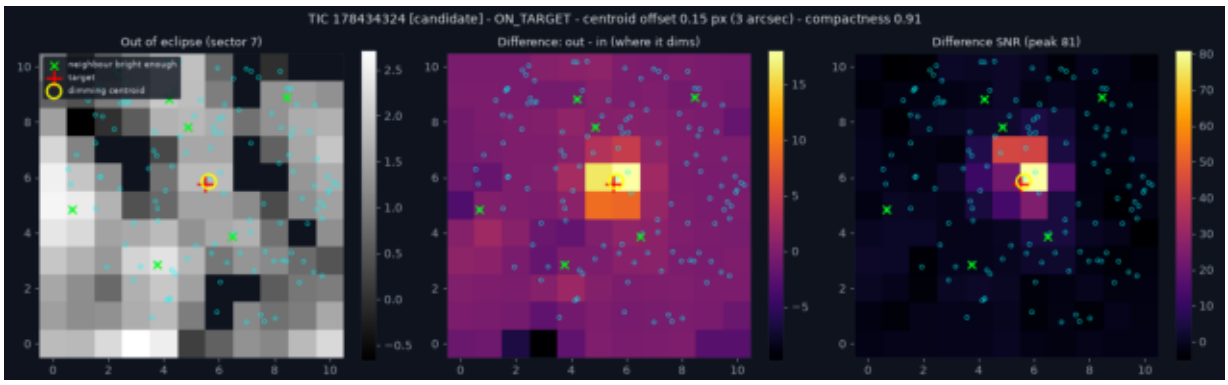
```
CATALOG CROSS-MATCH (novelty)

  VSX                                 no match
  ASAS-SN                             no match
  Gaia eclipsing-binary table         no match
  Gaia variable classification        no match
  TESS EB catalog (Prsa+2022)         no match
  ExoFOP TOI list                     no match

  Verdict: NOVEL
```

```
PIXEL-LEVEL SOURCE CHECK (TESScut difference imaging)
  Sector 7    light-curve dip in this sector: 165.0 sigma    t0 refined by 0.0000 phase
  Difference-image peak SNR   81.04   (gate >= 5.0)      Compactness   0.91   (gate >= 0.50)
  Dimming centroid offset     0.15 px (3 arcsec)   (gate <= 1.0 px, i.e. on the target)
  Nearest bright-enough neighbour in cutout: TIC 178434357
    2.12 px away, dTmag +1.40 - farther from the dimming than the target
```





No caveats - clean detection on every check.